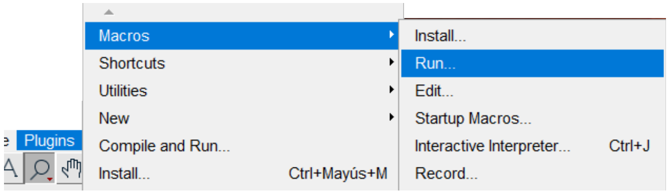
This macro requires a Fiji [1] installation and the following plugins:

- MorpholibJ [2]: https://imagej.net/plugins/morpholibj
- Labkit [3]: https://imagej.net/plugins/labkit/

Works with spatially calibrated RGB images stained with HDAB. Colour Deconvolution [4] plugin is used for DAB signal segmentation.

Running a Fiji/ImageJ Macro

1. Open Fiji
2. Go to **Plugins>Macros>Run…**
3. Select the macro **.ijm** file
4. Follow the instructions of the pop-up macro windows
5. **Once the image analysis is done, a new pop-up window will appear.**



Depending on the type of analysis other interactive windows will appear.

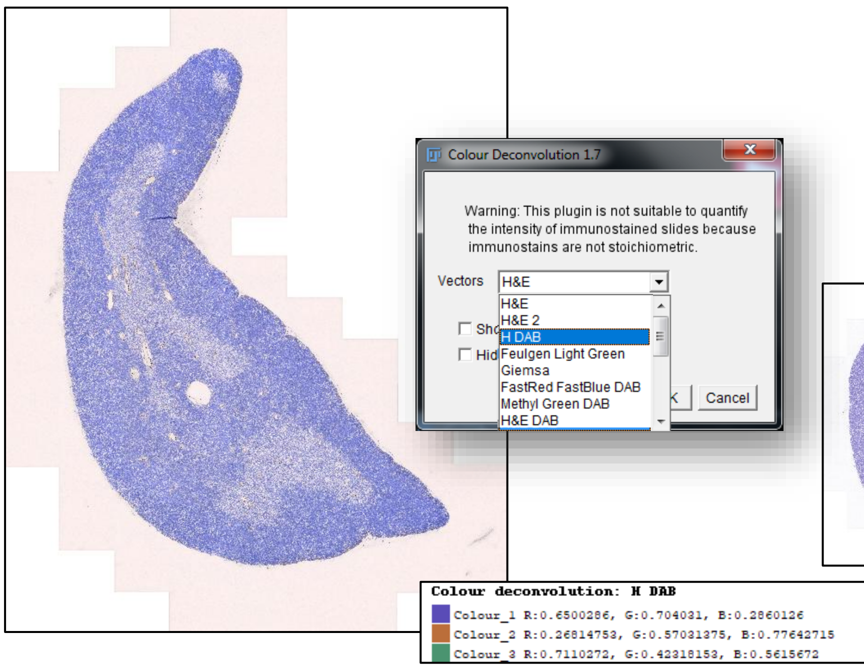
Follow their instruccions carefully and when done (only then!), press the OK button to continue with the analysis.

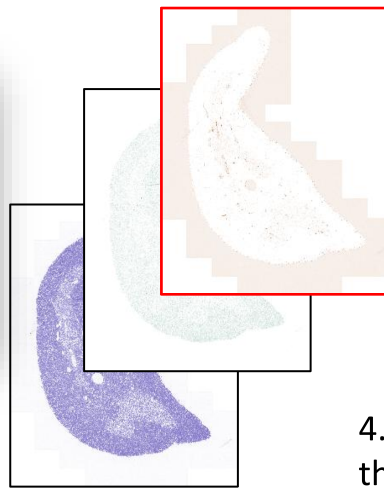
Cancel button will cancel macro execution.

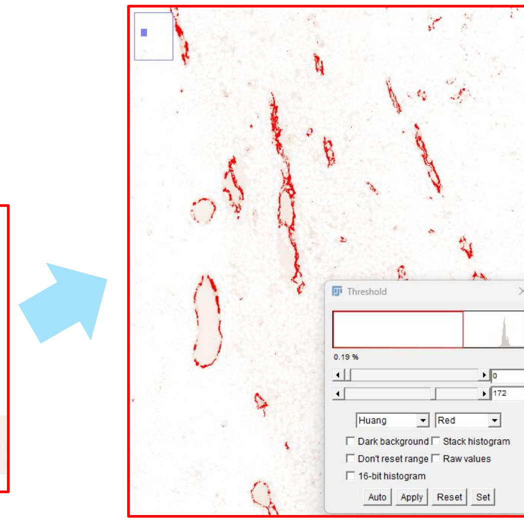
Colour Deconvolution Fiji Plugin

Uses different vectors to distinguish colour for different stainings.
Customized vectors can also be created selecting user defined areas
Splits images in 3 colors (dependent on vector used)

1. Go to **Image>Color>Colour Deconvolution**
2. Choose a Vector or create a customized one
3. After image has been splitted into 3 channels, an intensity Threshold (**Image>Adjust>Threshold**) can be used to select pixels with colour you want to analyze







4. To create a selection around thresholded pixels go to **Edit>Selection>Create Selection**
5. Measure area of Selection

Segmentation using Machine Learning Pixel Classification with Labkit plugin

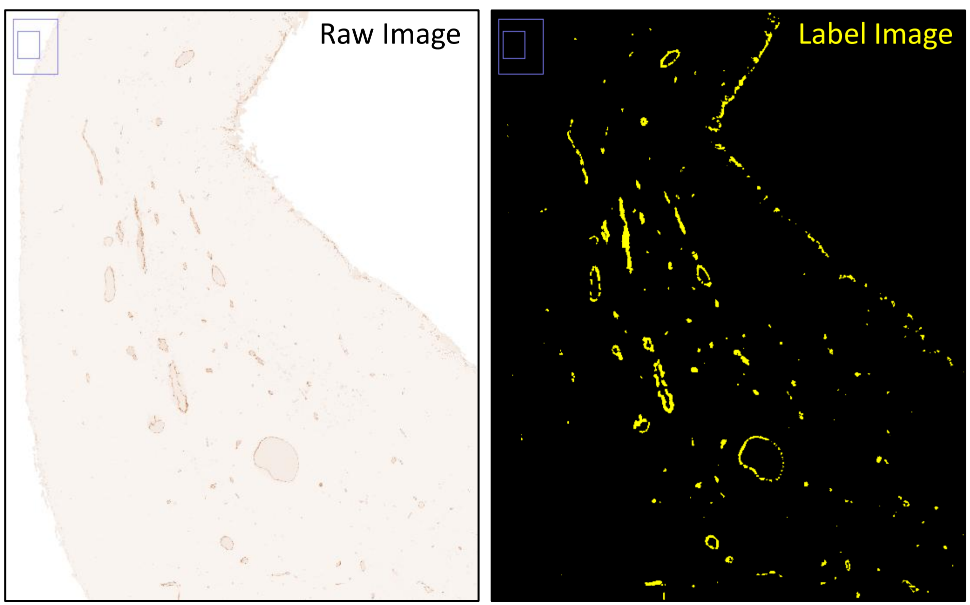
https://imagej.net/plugins/labkit/

https://imagej.net/plugins/labkit/gpu-accelleration-tutorial

Image pixels will be classified or divided into a user defined number of classes. Using drawing tools the user “shows” the software image pixels corresponding to each class (the so called “Labels”, and the process being “supervised training”). A set of features or filters can be used to train and create a classifier that will segment the image pixels into the especified classes. Once created, the same trained classifier is then used for segmentation of all the data set.

Labkit segmentation is based on a random forest pixel classification algorithm.

https://www.youtube.com/watch?v=6QSrgmMH4hE
https://imagej.net/plugins/labkit/pixel-classification-algorithm



In this case, a 2 classes classifier was created :

- Background/Tissue
- Vessels

Input image

- Macro will read and use image spatial calibration
- Tissue ROI creation and area measurement.
- Detection of medulla and cortex areas of tissue
  (user can edit medulla ROI if needed)
- Measure Area of Tissue, Medulla and Cortex ROIs

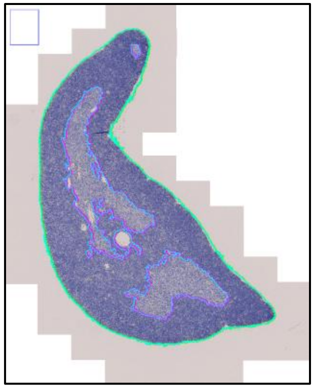
Colour Segmentation

- Use of Colour Deconvolution HDAB vector for vessel segmentation
- On the resultant colour window corresponding to DAB vessels, set threshold to exclude background and create positive signal selection. Then measure postitive DAB areas in previous ROIs.

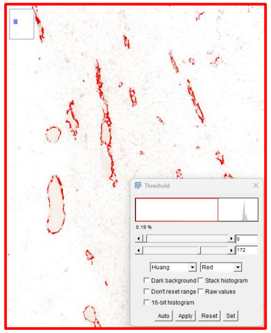
Vessel definition and segmentation using Labkit pixel classification

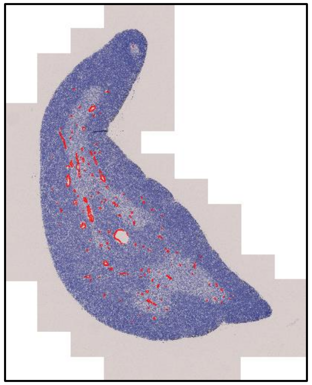
- Improved vessel segmentation on DAB colour image with Labkit
  (combined-vessels ROI can be edited by user if needed)
- Add Particles tools for isolated vessel fragments ROI creation
- Area measurement of individual vessels

Macro Pipeline

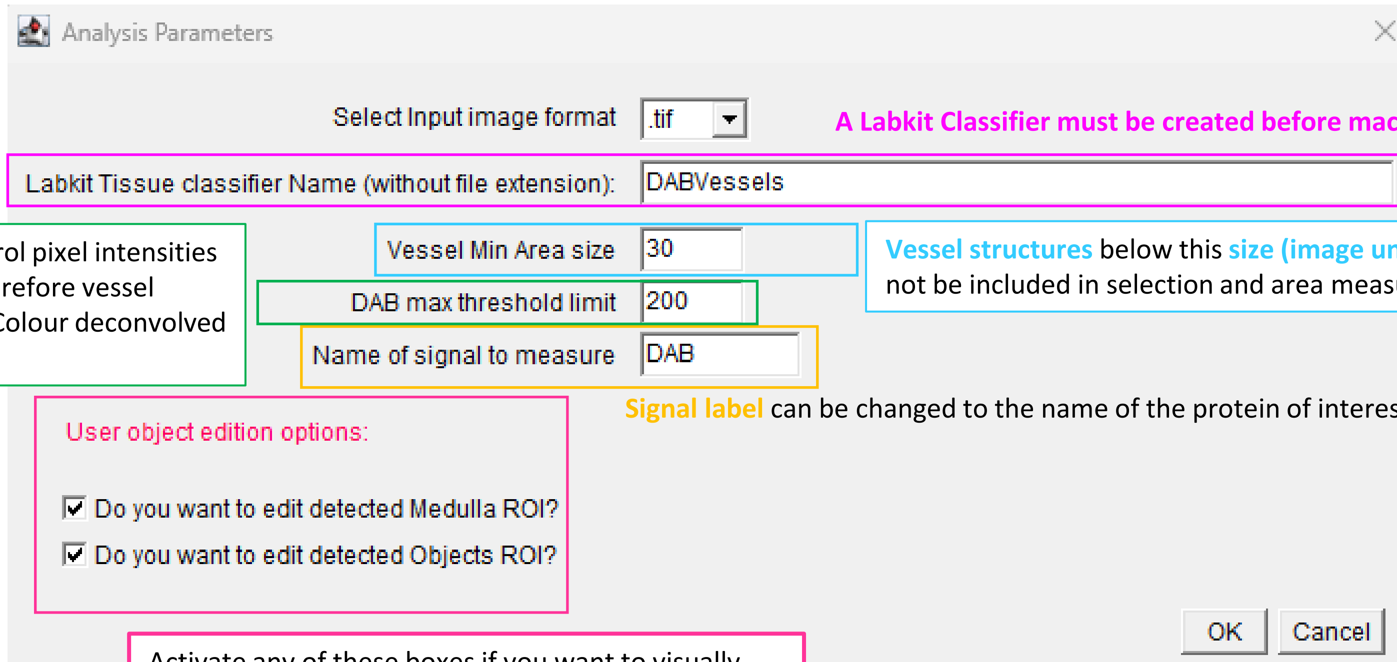


Tissue
Medulla
Cortex





Macro Dialog Box Options



A Labkit Classifier must be created before macro execution

Threshold to control pixel intensities detected, and therefore vessel segmentation, in Colour deconvolved image output

Vessel structures below this size (image units) will not be included in selection and area measurement

Signal label can be changed to the name of the protein of interest

Activate any of these boxes if you want to visually check and/or modify detected ROIs contours

Macro Outputs (in subfolder within input folder)

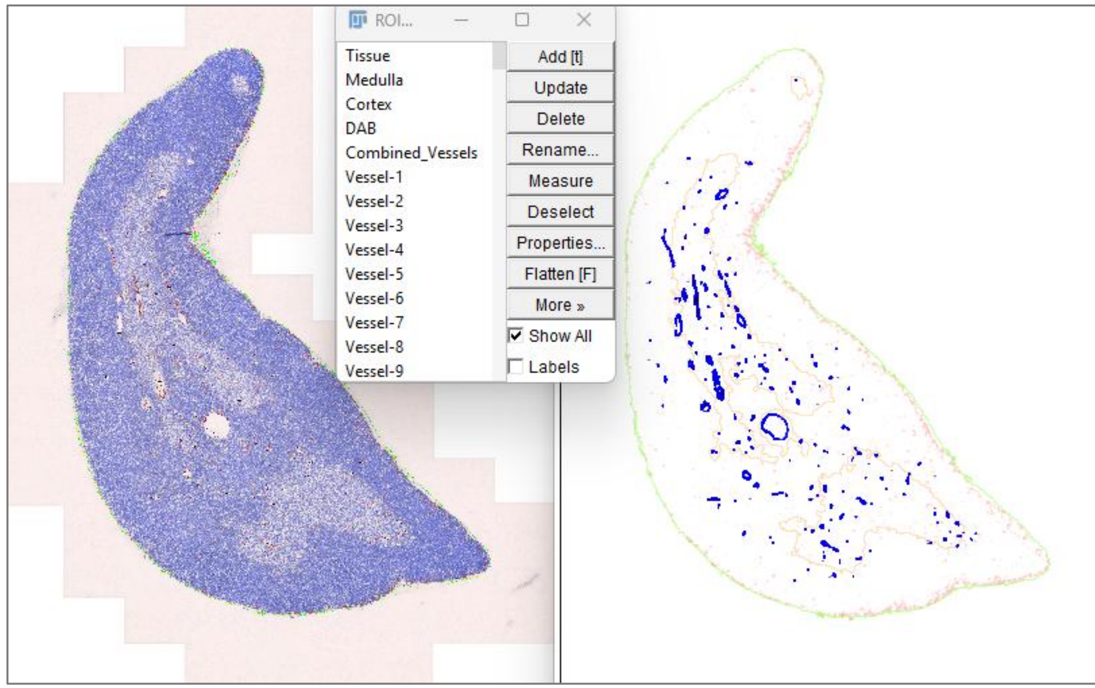
1. **ZIP ROI file.** One file per image analyzed.

2. **ROIs overlay and vessels binary images**. One of each per image analyzed.

3. **Individual vessel fragments area results table**. One per macro run.

4. **Summary results table**. One per macro run.

5. **Txt Log file**. One per macro run.



Vessel_Area_Results...

File Edit Font

	Vessel #	Area
1	1	43.318
2	2	81.699
3	3	51.304
4	4	60.452
5	5	826.672
6	6	32.912
7	7	50.336
8	8	137.746
9	9	40.946
10	10	107.835
11	11	54.063
12	12	50.239
13	13	433.035

Results

File Edit Font Results

	Image	Area Units (^2)	Tissue Area	DAB Area	% DAB in Tissue Area	Cortex Area	DAB in Cortex Area	% DAB in Cortex Area	Medulla Area	DAB in Medulla Area
1	_ __03_serie1.tif	microns	3241649.923	39436.910	1.217	2469264.823	23247.494	0.941	772385.100	16189.417

Citations

1. Schindelin, J., Arganda-Carreras, I., Frise, E. *et al.* Fiji: an open-source platform for biological-image analysis. *Nat Methods* **9**, 676–682 (2012). https://doi.org/10.1038/nmeth.2019

2. Legland, D., Arganda-Carreras, I., & Andrey, P. (2016). MorphoLibJ: integrated library and plugins for mathematical morphology with ImageJ. *Bioinformatics*, *32*(22), 3532–3534. doi:10.1093/bioinformatics/btw413

3. Arzt M, Deschamps J, Schmied C, Pietzsch T, Schmidt D, Tomancak P, Haase R and Jug F (2022) LABKIT: Labeling and Segmentation Toolkit for Big Image Data. Front. Comput. Sci. 4:777728. doi: 10.3389/fcomp.2022.777728

4. Landini G, Martinelli G, Piccinini F. Colour deconvolution: stain unmixing in histological imaging. Bioinformatics. 2021 Jun 16;37(10):1485-1487. doi: 10.1093/bioinformatics/btaa847. PMID: 32997742.